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RESEARCHMETHODS

Nature and Meaning of Research

Research is derived from two words Re and Search. Re means finding out again while Search means finding out as a result research simply mean finding out. The big question is what are we finding out? In research therefore we are interested in finding solution to problems.

Research is a systematic process used by scholars to provide solution to problems; to uncover facts in an attempt to formulate rules and generalization based on the facts uncovered through approved investigative procedures.

Research can be defined as a systematic way of finding out about the worth or otherwise of a given subject matter. Research has varied definitions depending on 'viewpoints and usage'. It is finding out the truth about an event or situation. The Webster dictionary defines research as a careful or diligent search, studious and critical inquiry and examination aimed at the discovery and interpretation of new knowledge.

According to Adewumi (1988) in Awoniyi, Aderanti and Tayo (2011), research is a process of finding out the aspect of knowledge or idea which people have not known about or they are not clear about them. It is also a process of finding out new information, new methods of doing things and an extension of boundary of knowledge. Osuala (1977) in Adewumi (1988) says research is simply the process of arriving at dependable solutions to problem through the planned and systematic collection and interpretation of data. Nwana (1985) describes research as a broad term which is a process of finding out the solution to a problem. The action word in this rather simple definition is the finding out or put in other words the 'searching for something.

It is clear from these selected descriptions that the existence of a problem or problems is a central core of what all researches is all about. No research can exist

in the absence of a problem. Identification of problems is therefore the beginning of such activities either singly or by groups of people.

In summary, research can be defined as an investigative process of finding reliable solution to a problem through a planned and systematic selection, collection, analysis and interpretation



of data relating to the problem.

What Research is not

1. Research is not simply rewriting other people's words or ideas into a neat description i.e. summarizing what is found in textbooks. It could only become a research when ideas are analyzed and organized into appropriate thought
2. Research is not a defense of one's convictions. Research seeks truth and as a result the position must be based on truth.
3. Research objective is to clearly present the truth and not to fight others position

Needs for Research

There are many reasons for carrying out research, but perhaps the most important is to broaden and deepen what you know by allowing you to become a mini-expert in your own area. Some of the needs for research are:

- Provide training in problem solving
- It allows one to become expert in his area
- Helps in testing what you think you know. Just because someone told you something how do you know it is true? Research can help you find out
- Helps in understanding the other side of an issue. Research can force you to look at an issue from more than one perspective
- There are quite a number of research oriented jobs for which knowledge of research can make one to be gainfully employed
- Desire to get a degree along with its consequential benefits
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RESEARCH PARADIGM, APPROACH AND METHODS

Paradigm

The term paradigm originated from the Greek word paradigm which means pattern and was first used by Thomas Kuhn (1962) to denote a conceptual framework shared by a community of scientists which provided them with a

convenient model for examining problems and finding solutions. Kuhn defines a paradigm as: “an integrated cluster of substantive concepts, variables and problems attached with corresponding methodological approaches and tools...” According to him, the term paradigm refers to a research culture with a set of beliefs, values,



and assumptions that a community of researchers has in common regarding the nature and conduct of research (Kuhn, 1977). A paradigm hence implies a pattern, structure and framework or system of scientific and academic ideas, values and assumptions.

During the past century, different paradigms have taken birth due to the remarkable growth in social sciences research.

There are mainly two paradigms:

Positivism and

Anti-positivism (or naturalistic inquiry)

Positivism

Positivist believes that true knowledge is based on experience of senses and can be obtained by observation and experiment. Positivistic thinkers adopt the scientific method as a means of knowledge generation

Positivistic paradigm thus systematizes the knowledge generation process with the help of *quantification*, which is essential to enhance precision in the description of parameters and the discernment of the relationship among them. Positivist paradigm adopts the quantitative approach.

Anti-positivism

Anti-positivism emphasizes that social reality is viewed and interpreted by the individual according to the ideological positions possessed. The anti-positivists believe that reality is multi-layered and complex and a single phenomenon is having multiple interpretations.

Anti-positivism involves probing into the various unexplored dimensions of a phenomenon rather than establishing specific relationship among the components. It emphasizes understanding and interpretation of phenomena and making meaning out of this process. It is basically qualitative in approach.

Research Approach

The research method is a strategy of enquiry, which moves from the underlying assumptions to research design, and data collection (Myers, 2009). Although there are other distinctions in the research modes, the most common classification of research methods is into qualitative



and quantitative. The terms refer to research methods, that is, the way in which data are collected and analyzed, and the type of generalizations and representations derived from the data.

Research Design

Research Design can be thought of as the logic or master plan of a research that throws light on how the study is to be conducted. It shows how all of the major parts of the research study– the samples or groups, measures, treatments or programs, etc– work together in an attempt to address the research questions. Research design is similar to an architectural outline.

The research design refers to the overall strategy that you choose to integrate the different components of the study in a coherent and logical way, thereby, ensuring you will effectively address the research problem; it constitutes the blueprint for the collection, measurement, and analysis of data

A research design is the *set of methods* and procedures used in collecting and analyzing measures of the variables specified in the problem research.

Table 1: Selection of research paradigms and research methods

Research paradigms	Research approach	Research methods	Examples
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Positivism	Quantitative	Surveys	- Attitude of distance learners towards online based education
		Correlational	- Relationship between students' motivation and their academic achievement.
		Experimental	- Effect of intelligence on the academic performances of primary school learners
		Quasi-experimental and ex-post facto research	The effect of Grade Zero attendance on achievement at the end of the first grade

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Anti-positivism	Qualitative	Biographical	- A study of autobiography of a great statesman.
		Phenomenological	- A study of dropout among the female students
		Ethnographical	- A study of the culture and educational practices among the Yorubas of Nigeria
		Case study	- A case study of an open distance learning Institution in a country.
Critical theory	Critical and action-oriented	Ideology critique	- A study of development of education during the British rule in India
		Action research	- Absenteeism among standard five students of a primary school

CLASSIFICATION OF RESEARCH

Research could be classified based on the following criteria:

- i. Classification by purpose
- ii. Classification by method.
- iii. Classification based on type of evidences required and method of analysis.

Classification of Research by Purpose

Classification of research by purpose is based primarily on the degree to which findings have direct application in education or the degree to which they are generalizable to other educational situations. Types of research based on purpose or use are:

- Basic (Pure) research

- Applied research.
- Action research

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- Evaluation research
- Research and Development (R & D)

Basic [Pure] Research: This is concerned with the production of result and finding which lead to development of theory. This type of research is not for finding solution to problem but rather interested in building theories which can lead to advancement of knowledge in a particular field. For example, much basic research has been conducted with animals to determine principles of reinforcement.

Applied Research: As the name implies, it is conducted for the purpose of applying or testing theory. It is concerned with the usefulness of ideas or theories to practical situation. Applied research provides data to support theory, guide theory revision or suggest development of new theory. For example which forms of motivation brings about better academic performance among students?

Action Research: This type of research is directed towards solving specific problem in a local setting. It is usually taken by a teacher or group of teachers in school to solve specific class room problem so that the result can be generalized. The primary goal of action research is to provide solution to a given problem not to contribute to knowledge. It is meant to provide immediate answers to problems that cannot wait for critical solutions. For example, repeating a lesson after realizing that students do not have a clear understanding of the topic.

Evaluation Research: The results of evaluation research help in decision making. It helps to decide whether a programme is to be continued, modified or be stopped. Evaluation is traditionally classified according to its purpose namely formative and summative evaluation.

Formative evaluation is used for the purpose of program improvement. It is carried out when a programme is in progress.

Summative evaluation is used for the purpose of making summary judgments about a program and decisions to continue or discontinue the program.

A newer and current popular way to classify evaluation research is to divide it into

five types Needs assessment, which asks this question, is there a need for this type of program?

Theory assessment, which asks this question, is this program conceptualized in a way that it should work?



Implementation assessment, which asks: Was this program implemented properly and according to the program plan?

Impact assessment, which asks: Did this program have an impact on its intended targets? Efficiency assessment, which asks: Is this program cost effective?

Research and Development: This is aimed at developing and testing product to ensure their effectiveness. For example, the development of teaching and learning aids to improve assimilation of content by pupils/students.

Classification of Research by Method

This classification is based on the method or strategy to be used in the conduct of a research. All methods have certain procedure in common such as statement of a problem, collection of data, analysis of data and drawing conclusions. Classification by method appears to be the most efficient as it minimizes categories and maximizes differentiation. It places all research into five categories namely.

- Historical
- Descriptive (survey)
- Correlational
- Experimental and
- Casual-comparative (ex-post-facto)

Historical Research

This type of research is based on oral evidence, records such as diaries, case history, autobiography, logbook; Books, Journals, magazines etc. The purpose of an historical research is to obtain a better understanding of the present through the evaluation of the past and intelligent prediction of the future. An historical research aids in avoiding past errors and predicting the future. It also refreshes ones memory on what is known and unfold what is not known.

There are two main sources of collecting historical data. They are primary and secondary sources. Primary sources are relics and other things that have direct physical relationship e.g. observation, participation etc. A secondary source deals with bibliographies,

references, materials and documents recorded by someone else. These are less reliable than

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primary sources. In this type of research approach statistical hypotheses are rarely used. Example: Historical Development of Bugema University.

Descriptive Research

It is also called a survey research. This type of research is based on information, interviews, (oral, written, structured, unstructured etc), inventories, rating scales, self-report, observation etc. Descriptive research is aimed at fact finding of the present condition or current situation. The result obtained through this procedure can be statistically analyzed. A descriptive survey generally tries to collect information from a representative group and based on such information inferences are drawn about the behavior of the entire population. The descriptive survey is usually prompted by the need to know what the current situation is with regard to a particular educational planning problem, but it may not tell the researcher what has caused the problem nor will it supply alternative solutions.

The major limitations of this research approach is found in the questions asked as people may not actually behave the way they respond to questions. Moreover, the descriptive survey also provides no indication of the relationship between the variables studied. They only provide information without explanation.

Example of Descriptive survey topics

- (a) A survey of decision-making practices of secondary school principals.
- (b) A study of university students' attitude to online registration.
- (c) A study of factors affecting the teaching of Mathematics.

Correlational Research

The correlational studies present a research approach that seeks to explain educational phenomena rather than mere stating the phenomena. It also requires a representative sample, but in this case it demands much more sophisticated

statistical analysis to be carried out. A correlational study provides evidence of relationship. The major advantage of correlational studies is that relationships are measured in their real setting. A major disadvantage of this



approach is that since real life is complicated attempt to control certain variables and examine one or two aspects of behavior in isolation may be difficult.

The following are examples of typical correlational studies:

1. *The relationship between intelligence and academic performance. Scores on an Intelligence test and performance test would be acquired from each member of a given group. The two sets of scores would be correlated and the resulting coefficient would indicate the degree of relationship.*
- 2.. *Use of entry grades to predict success at university. Entry grade point would be correlated with ultimate Cumulative Grade Point Average (CGPA) at the end of university education. If the resulting coefficient is high, the entry grade would be considered a good predictor.*

Experimental Research

This type of research involves conducting experiment for research purpose. The experimental method involves investigating possible cause and effect relationship by exposing one or more experimental groups to a treatment and one or more control groups not receiving the treatment

The simplest form of experiment involves only two measures the dependent variables otherwise regarded as “effects” and the independent variables, otherwise known as causes. In an experiment two or more groups are necessary. These include a control group and at least one experimental group.

To enable one to compare, the group must be as near identical as possible. The main problem faced in carrying out experimental research in education is that in trying to control the independent variable and setting up matched groups, one may be interfering with normal school process. This is why there are limitations to adoption of experimental research in education, since we cannot experiment on people in a way which may affect their lives adversely.

Example of typical experimental studies:

1. The comparative effectiveness of discussion and lecture method on pupils' academic



performance. The independent variable or “cause” is type of instruction (discussion versus lecture); the dependent variable or “effect” is academic performance. Two groups (preferably randomly formed) would be exposed to essentially the same experience except for method of instruction. After some period of time, their academic performance would be compared.

Causal-Comparative Research

Causal-comparative and experimental research represent distinctly different methods, however both attempt to establish cause-effect relationship, and both involve group comparison. In causal-comparative study the independent variable or ‘cause’ is not manipulated, it has already occurred. Independent variables in causal-comparative studies are variables which cannot be manipulated e.g. gender, (male -female), location (urban-rural).

The following are examples of typical causal-comparative studies

1. *The effect of Grade Zero attendance on achievement at the end of the first grade.* The independent variable or cause is *Grade Zero attendance* (students attended Grade Zero or they did not); the dependent variable or effect is *the achievement at the end of the first grade*. Two groups of first graders would be identified - one group which had attended Grade Zero, and one which had not. The achievement of the two groups would be compared.
2. *The effect of gender on academic achievement of secondary school students.* The independent variable or cause is *gender (male and female)*; the dependent variable or effect is *academic achievement*. The achievement of males would be compared to the achievement of females.

Developmental Research

Developmental researches want actually to study the growth characteristics of a child. These growth characteristics include intellectual, emotional and physical. It aims at measuring growth of all these over time.

Developmental research in Education involves a number of specific tasks of study

1. Longitudinal studies
2. Cross-sectional studies

3. Trend/predictive studies

All of these measure development over time.



Longitudinal Studies

This term is used to describe a type of study conducted over a period of time which could be long or short. Data concerning the subject to be studied are gathered over a period of time. A short longitudinal study may take a week, while a long one might take years even up to fifty years. In longitudinal studies, the sample is constant throughout the study. This distinguishes longitudinal studies from other types of developmental research. Depending on the time, data are collected at intervals on the intended characteristics.

There are two basic types of longitudinal studies;

- i. **Prospective longitudinal studies.** Here the emphasis is beginning from a point and follow the sample into the future. The measurement is done into an established date in future.
- ii. **Retrospective longitudinal studies.** Here the emphasis is to establish a point and then go into the past of the particular sample. See what happen to the sample in the past. This type is common in the area of psychology and Guidance and Counseling. Looking into the past to know what contributed to their present state. Retrospective longitudinal studies are less reliable as one need to collect data from those associated with the sample.

Advantages of Longitudinal Studies

- i. The longitudinal studies unlike other type of developmental research is able to identify typical patterns of development and find out factors that determine development in individual. It enables one to establish general relation. It enables the researcher to examine individual variation in characteristics or traits and produce growth curves for development of individual and rate of growth.
- ii. Time which is usually a limiting factor in research is generally more readily available in longitudinal studies.

Disadvantages

- i They are time consuming and resource consuming and as a result only rich researchers can carry out longitudinal studies, as samples has to be transferred wherever there is cause for movement.
- ii. Sample mortality: There is probability of death of some of the subjects no

matter how short the time might be and depending on the size of the sample. Once sample die one cannot choose another sample. The choice of a new sample will affect the result of the research.

iii They impose considerable problems of organization on the researcher e.g. Keeping track on the sample or keeping contact with the sample. Longitudinal studies are the most important of all development researches.



Cross - Sectional Studies

Cross-sectional study is one in which different respondents are studied at different point in time. A cross-sectional study is often referred as a "snapshot" of a population at a particular point in time. Here the samples are not constant and we are not concerned about an individual. On the basis of cross-sectional one will be able to determine development at different ages or time.

A major advantage of Cross-sectional studies over longitudinal studies is that of time. While longitudinal studies will study over several years, cross-sectional study might be concluded within a year. Longitudinal studies are more reliable as the same sample is been used and might not be affected by circumstance as when different samples are used.

Trend/Predictive Studies

The trend/predictive studies attempt to examine recorded data to establish pattern of changes. With the past trend this type of study can predict into the future. The trend/predictive studies is not limited to individual or sample alone but can be used even for the development of a population.

Advantages

- i. More economical in term of time.
- ii. Bring quick results.

Disadvantages

- i. Reasonable sample needed.
- ii. Not easy to draw conclusion as circumstances might change

VARIABLES IN RESEARCH

The critical factors of an investigation are known as variable. A variable is anything that is changeable or alterable. That is, a characteristic that take on different conditions e.g. attitude, skill, knowledge, interest, weather, time of day etc. In its specific research meaning, a variable refers to a characteristic of group of objects, people, events or situation which may change or differ.

The most frequently encountered variables in research are independent variable,

dependent variable, extraneous variable, moderator variable and control variable.

a Independent or treatment variable



This is the variable that determines the nature and/or occurrence of another variable. Sometimes, this variable is called a predictor variable because it is responsible for the outcome a researcher get. For example a researcher may want to see the effect of interest on achievement in mathematics. The independent variable here is interest.

b. Dependent variable

This is sometimes called criterion variable or the effect. This type depends on the treatment and can be manipulated. In the example above, achievement scores in mathematics are related to the independent variable (interest).

c. Extraneous (intervening variable)

This is a variable which may or may not be known, but which in some way affects the outcome of a research study. For example fatigue, hunger or time of the day may affect the outcome of a lesson or text.

d. Moderator

These are secondary independent variables which may not have been 'taken into consideration in a research study. Again, using the example of achievement in mathematics, it is possible that difference in gender, aptitude or socio-economic background of the subject might affect subjects' achievement in mathematics.

Classification Based on Methods of Data Analysis.

There are three types of research based on methods of data analysis. These are:

- i. Qualitative - non computational
- ii. Quantitative - Computational
- iii. Triangulation or multiple perspectives.

Qualitative Research

Qualitative research has to do with collecting, analyzing, and interpreting data by observing what people do and say. This type of research is less objective than quantitative research and uses different methods of collecting information, mainly individual, in-depth interviews and focus group.



Qualitative Research Types

- *Phenomenology*
- *Ethnography*
- *Grounded theory*
- *Historical research*
- *Case study research*

Phenomenology – a form of qualitative research in which the researcher attempts to understand how one or more individuals experience a phenomenon. For example, you might interview 20 widowers and ask them to describe their experiences of the deaths of their wives.

Ethnography – is the form of qualitative research that focuses on describing the culture of a group of people. Note that a *culture* is the shared attitudes, values, norms, practices, language, and material things of a group of people. For an example of ethnography, you might decide to go and live in Yoruba communities and study the culture and their educational practices.

Grounded theory – is a qualitative approach to generating and developing a theory from data that the researcher collects. For an example, you might collect data from parents who have pulled their children out of public schools and develop a theory to explain how and why this phenomenon occurs, ultimately developing a theory of school pull-out.

Historical research – This is research about events that occurred in the past. An example, you might study the use of corporal punishment in schools in the 19th century.

Case study research – is a form of qualitative research that is focused on providing a detailed account of one or more cases. For an example, you might study a classroom that was given a new curriculum for technology use.

Quantitative Research

Quantitative research is research involving the use of structured questions where response options have been predetermined and a large number of respondents are involved. In this type of research, measurement must be objective, quantitative and statistically valid. It is all about numbers and objective hard data.

Common differences usually cited between these two types of research include

Qualitative	Quantitative
Subjective	Objective
Language emotions	Language numbers
Method of investigation include focus group discussions, one-on-one interviews, write-in response, bibliographic studies, case studies	Statistical survey like Likert multiple choice, t scale etc.
Research questions only	Developed from research questions, null hypothesis may be statistically tested
No statistical test	Statistical tests possible
Exact replication not possible	Replication possible
Potential dependability varies but usually low	Potential dependability varies but usually high
Literature review may be done as study progresses or afterwards	Literature review must be done early in the study

Triangulation or multiple perspectives (Mixed Methodology)

This involves the merging of qualitative and quantitative methodologies. Triangulation is the term used to describe the combination of several qualitative methods or combining qualitative with quantitative methods. Four strategies for combining methodologies are common. These are;

Qualitative and quantitative studies are conducted simultaneously.

A qualitative study can be ongoing while multiple waves of quantitative studies are done

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A qualitative study can precede a quantitative study. A quantitative study can precede a qualitative study.

THE RESEARCH PROCEDURE OR TECHNIQUES

The research procedure is nothing other than a sequence of activities which when followed, will enable the investigator to achieve his aim. The following sequence or step is recommended if one is to embark upon a programme of research. These basic steps include:

1. Finding a suitable research topic or identifying problem,
2. Defining the problem clearly,
3. Reviewing available relevant literature to the problem.
4. Formulating hypotheses
5. Defining the population and selecting sample.
6. Designing and validating the instruments for data collection
7. Collecting the data
8. Organization of data.
9. Analysis of data
10. Interpretation of data
11. Conclusions
12. Writing the report.

The Research Problem

There is no doubt that finding a suitable research topic or research problem

requires a lot of guidance and deep thinking. The choice of an area for research is a personal decision, depending mainly on the interest and attitude of the researcher. For students to be able to locate and choose suitable research topic the following guides may be used.



- a) Personal motivation and interest in the study.
- b) Usefulness to the society.
- c) Contribution to knowledge.
- d) Time available for the study.

Ideas for a good research topic may come from personal observation and experience or from review of literature, and the choice of topic depend largely on interest and availability of facilities for the study.

Sources of Research Problem

Research problem can be found from a number of sources such as

1. Unanswered questions: Past researcher may generate some questions which he fails to answer in his research.
2. Literature Review: In reviewing literature, examine suggestions for further researches at the end of dissertations, in critique of other people research especially in review.
3. Unsatisfactory state of affairs: If people feel unsatisfied with the way things are it is an indication that something is wrong, which has to be rectified. This may be source of research problem.
4. Missing link: A gap in thought process or practice e.g. what motivates students?
5. Replication of study: Repeating a research within the same context after a period of time or a study repeated under different geographical location.
6. Quasi-replication – previous topic repeated with alteration to some part of the design
7. Personal experience: Through careful observation of practice in the area of one's interest, a researcher can identified researchable problems.

8. From Existing Theories: Most theories are good sources of researchable problem because theories normally give rise to deduction, which need to be tested through the instrumentality of research.



9. Original idea – develop a truly researchable topic after extensive literature review
10. Recommendations from your professor or advisor

Research Topics to be avoided

- Avoid philosophical topics or those based on personal belief
- Avoid strictly bibliographic topics that are already discussed fully in book-length studies
- Avoid describing processes. Such information will not allow for original insight and judgment on your part.
- Avoid topics too narrow or too recent for discussion. It will not allow for adequate use of research materials.
- Avoid philosophical topics or those based on personal belief
- Avoid strictly bibliographic topics that are already discussed fully in book-length studies
- Avoid describing processes. Such information will not allow for original insight and judgment on your part.
- Avoid topics too narrow or too recent for discussion. It will not allow for adequate use of research materials.
- Topics that are too big
- Topics that can be traced to a single source.
- Topics that are trivial – of small value
- Topics that are overused (over researched)

REVIEW OF AVAILABLE RELEVANT LITERATURE

A major step in conducting research is literature review. After selecting a topic one should make a thorough survey of what had already been known in the area of interest. Literature review is an examination and gathering of valuable topical and pertinent information already expressed from sources on issues of study which tend to modify and add weight to your study. It implies knowledge which is relevant and can buttress as well as negate the arguments of study in order to make contribution to your field of study.

Also, it is through a review of previous work and findings in the area chosen that enable the researcher to develop his own theoretical or conceptual framework upon which hypotheses



are generated.

The sources of literature to be reviewed on any problem include Journals, research review, encyclopedia, textbook, monographs, periodicals, research report, newspaper, bulleting, theses, dissertation, seminar paper etc.

The presentation of literature review should be systematic and should cover all the variables involve in the study. A brief summary of the literature reviewed should be included in this aspect of the research report.

Roles of Literature Review

- a. A thorough knowledge of related research work enables the investigator to defined frontiers (boundaries).
- b. Literature review enables researchers to place their questions in the right perspective.
- c. By studying related research, the investigator learn which procedure and instruments have proved useful and which ones seem less promising and had to be improved upon;
- d. A thorough search through previous research works makes an investigator to avoid unintentional replication of previous studies.
- e. The study of related literature places a researcher in a better position to interpret the significance of their own result.
- f. Literature review helps in surveying what is known and what is not yet known;
- g. Literature review helps to identify previous research procedure instruments used, utility of the instruments, how to adapt or adopt the instruments, research design and hypotheses investigated.
- h. Literature review helps to interpret findings and integrate them into the main body of knowledge.

Procedure for Literature Review

- a. Identify appropriate literature to be thoroughly studied;
- b. Read the abstract or summary section of the literature to determine

relevance and appropriateness;

- c. Skim the report to determine relevance before going on its development.
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waste your time reviewing irrelevant literature.

d. Begin with most recent studies in your area of interest/study and then work backward through earlier studies

e. Read-widely in the selected textbooks/ journals, manuals, archives-letters, minutes, diaries, biographies, autobiographies monographs, etc.

f. Be sure you indicate which parts of the notes are direct quotations from the author, and which ones are your own paraphrases.

g. Literature review should be thorough and complete. A researcher should not rush research until after a thorough review of literature.

i. Your work should highlight relevance of the findings in the previous studies, gap of knowledge to be contributed by your own study.

k. Avoid replication of work done as discovered in the literature review,

Research Objectives

Objectives are statement of intent, a statement which indicates what the researcher aim to achieve at the end of the research. Objectives must always be set after having formulated a good research question. After all, they are to explain the way in which such question is going to be answered.

Research is an organized investigation of a problem in which there is an attempt to gain solution to problem.

To get right solution of a problem, clearly defined objectives are very important. Clearly defined objectives enlighten the way in which the researcher has to proceed. A research objective is a clear, concise declarative statement, which provides direction to investigate the variables.

Generally research objectives focus on the way to measure the variables, such as to identify or describe them. Sometimes objectives are directed towards identifying the relationship or differences between two variables. Research objectives are the results sought by the researcher at the end of the research. It summarizes what is to be achieved by the study. Objective should therefore be closely related to the statement of the problem.

Objectives help the researcher to be focused on the study to avoid the collection of unnecessary data and as well as organize the study in clearly defined parts or phases. Clearly



defined objectives will facilitate the development of research methodology and will help to orient the collection, analysis, interpretation and utilization of data.

Objectives are usually headed by active verbs such as: to identify, to establish to describe to determine, to estimate, to develop, to compare, to analyze etc.

For example, the research objectives for a study of rate of unemployment in a particular region may be:

1. To compare the unemployment rate among all countries in the region.
2. To analyze the unemployment rate evolution from 2009 to 2019.
3. To identify the factors associated with high unemployment rates.

An objective is a clear statement of something that needs to be accomplished over a period of time and therefore must be SMART. SMART objectives are:

Specific – states exactly what you need to achieve

Measurable – includes a quality or quantity measure

Attainable – must be achieved

Realistic – can be challenging but must be practicable and credible

Time bound – with a clear end date or timescale. For example, to increase Bugema University enrolment by 10% is not time bound, while to increase Bugema University enrolment by 10% over the next six month is time bound.

Research Questions

Developing a good research question is one of the first critical steps in the research process. The research question, when appropriately written, will guide the research project and assist in the construction of a logical argument. The research question should be a clear, focused question that summarizes the issue that the researcher will investigate. Research questions are to be based on the objectives of the study.

How to Develop a Good Research Question:

- Researchers should begin by identifying a broader subject of interest that lends itself

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to investigation. For example, a researcher may be interested in unemployment

- The next step is to do preliminary research on the general topic to find out what research has already been done and what literature already exists. How much research has been done on unemployment?
- Then begin to narrow the topic by asking open-ended "how" and "why" questions. For example, a researcher may want to consider the factors that are contributing to unemployment. Create a list of potential questions for consideration and choose one that interests you and provides an opportunity for exploration.
- Finally, evaluate the question by using the following list of guidelines:
 - Is the research question one that is of interest to the researcher?
 - Is the research question researchable? Consider the available time frame and the required resources. Is the methodology to conduct the research feasible?
 - Is the research question measurable and will the process produce data that can be supported or contradicted?
 - Is the research question too broad or too narrow?

The following research questions could be derived from the objectives stated above.

1. Is there any difference in the unemployment rate among countries in the region?
2. What is the unemployment rate from 2009 to 2019?
3. What are the factors associated with high unemployment rates?

Formulating Hypotheses

Hypothesis could be defined as a prediction of the outcome of a research problem.

It is a suggested answer to the research problem. It is always put in a declarative statement to relate the variables to each other generally or specifically.

Two types of hypotheses are:

Null hypothesis

Positive

hypothesis

Null hypothesis is stated in the negative form e.g. there is no significant relationship between

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sex and performance in mathematics.

The positive (alternative) hypothesis is stated positively e.g. there is a significant relationship between sex and performance in mathematics.

Alternative hypotheses could be directional or non-directional

It is not in all research that we formulate hypotheses in some we have only research questions.

Attributes of good Research Hypotheses

- Hypotheses should be correctly, clearly and precisely stated.
- Hypotheses should be testable and measurable.
- Hypotheses should state relationships between variables.
- Hypotheses should be limited in scope.
- Hypotheses should be stated as far as possible in simple terms

Theoretical and Conceptual Framework

Some researchers use both the theoretical and conceptual framework; it is advisable to use only one. They both provide clear explanations regarding the relationships of the variables

Theoretical Framework

This involves presentation of different theories showing the generalizations already expressed in relation to the problem.

Conceptual Framework

Conceptual framework is a detailed presentation of the variables to be observed in a study. Topic: Challenges in the Implementation of Inclusive Education.

Conceptual Framework

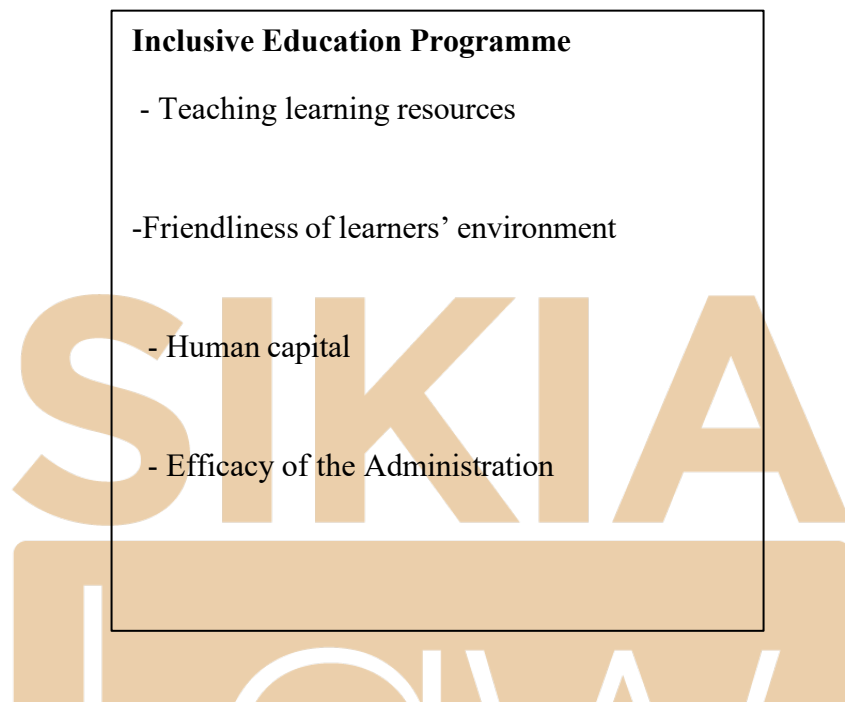
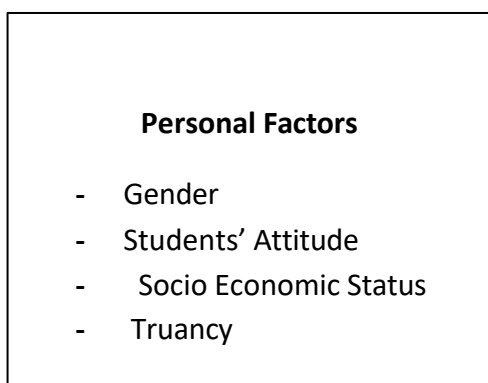


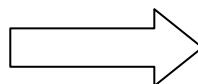
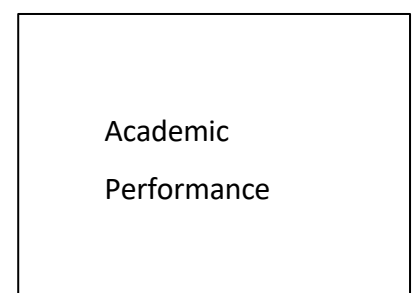
Figure 1: Variables Associated with the Study

Topic: Influence of Selected Personal factors on Students' Academic Performance

Independent Variables



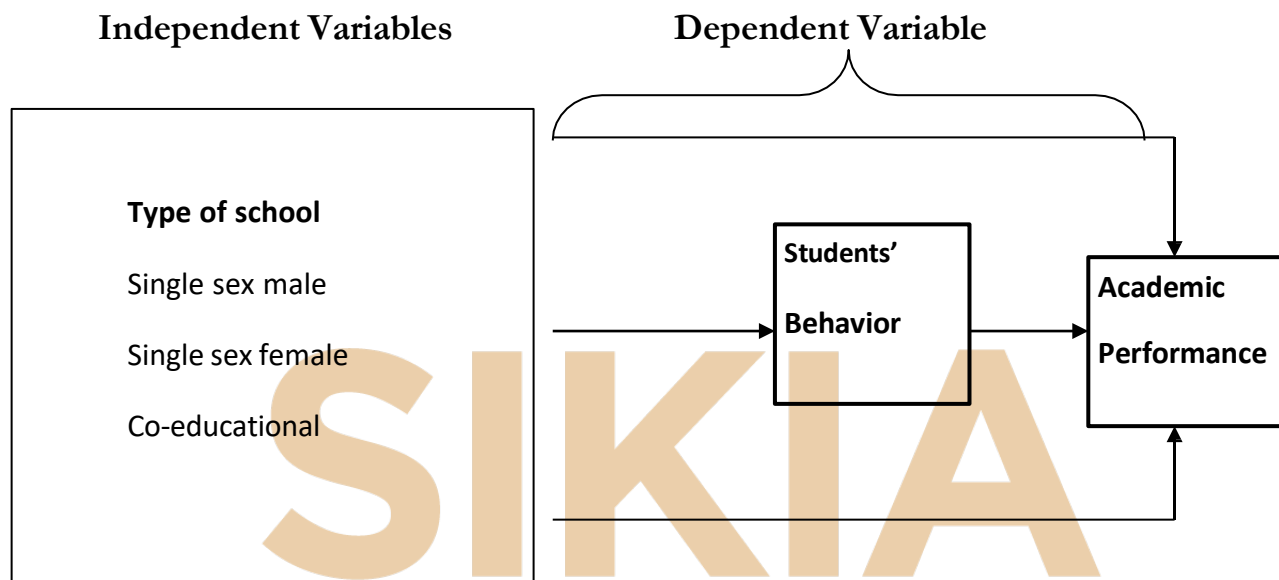
Dependent Variable



Topic: Comparative Analysis of the Effect of School Type on Secondary School

Students' Behavior and Academic Performance

Conceptual Framework



Sample and Sampling

Technique Defining the Population and Selecting Samples

A population is defined as "all members of any well-defined class of people, events of objects". The first step in selecting the number required from the group of the people or objects is to define the population. This enable the researcher to establish clear boundaries of the conditions which specifies who shall be included or excluded from the study.

Selecting Samples

A sample is a portion of a population. To study a population, it is often necessary to take a representative sample. The sample means selecting some members of the entire population for use in the study. The main reason for studying a sample is to be able to make generalized statements about the entire population. To study the

entire population may be difficult, time consuming and of course very costly, hence sample takes a fair portion as representative of the entire population.



There are certain generally accepted methods of sampling which may be used. These are:

Random sampling (probability sampling)

- a. Simple random sampling (Balloting or Dip-hand method),
- b. Systematic sampling.
- c. Stratified random sampling.

Non-probability sampling

- a. Purposive sampling.
- b. Cluster sampling
- c. Captive sampling.
- d. Convenience sampling.
- e. Voluntary sampling.
- f. Snow ball sampling.
- g. Event sampling.
- h. Time sampling.

Simple Random Sampling

This is the best known of the sampling procedures. The basis characteristic is that all members of the population have an equal chance of being included in the sample. For example if a population consists of 300 members the chance of selecting each member is $1/300$.

The steps in simple random sampling are:

- i Define the population.

- ii List all members of the population.
- iii Select the samples by employing a procedure where sheer chances determine which



member on the list are drawn from the population.

Simple random sampling can be carried out by balloting, card reshuffling, and use of table of random digit.

Balloting: in balloting name or number is assigned to each element in the study by writing the name or number on pieces of paper, squeeze the paper and start picking it one after the other until the researcher have picked enough number he wishes to use as sample.

Card reshuffling: like balloting name and number is assigned to each element and then the name or number is writing on each card to represent an element in the study. These cards are put together and shuffled. At intervals or at each time the cards are shuffled the researcher picks the card on top until he has picked enough to make up the sample.

Use of Table of Random Digits: This involves numbering all the elements in the population. Then select digits from the table in any systematic way and the element whose number coincides with the random digits are included in the sample this process is continued until the desired sample size is reached.

Systematic Random Sampling

This involves collecting the list of the study population and then choose sample in a systematic way. This procedure involves drawing a sample by taking every Kth case from a population. Systematic sampling is also referred to as interval sampling.

For example: if a sample of 50 students were to be selected from a departmental list with 500 names, the list of students in alphabetical order will be taken and every 10th name will be chosen as an elements in the sample.

Systematic sampling differs from simple random sampling in that various choices are not dependent. Once the first case is chosen, all subsequent cases to be included in the sample are automatically determined.

Stratified Random Sampling

When the population consists of a number of sub group or strata that may differ in the characteristics being studied, it is often desirable to use a form of sampling technique called stratified random sampling. This procedure can be divided into proportionate and



disproportionate. In proportionate stratified random sampling, sample are pick according to the number of subject in each strata while in disproportionate the strata are not equally represented it is either over represented or under represented.

Purposive) Sampling

The deliberate selection technique is used when there are specific cases or members of the population with certain crucial traits to the study and the researcher must find a way of including them. Sample that best meet your objectives or your target group.

Cluster Sampling

The cluster sampling is fairly similar to the stratified random sampling technique Cluster sampling is otherwise known as multistage sampling in that the population is sub-divided into units and a random sampling of smaller units that is, it is a population that exist in cluster over a geographical area. The selection of individual case in the group may also employ simple random technique. This method is used when the population is very large and covers a large geographical area. For example a country can be divided into regions or states which constitute sampling unit called clusters.

Captive Sampling: The captive sampling is employed when the researcher decide to use only the sample he can get. This is what television Reporters use in sampling public opinion on topical issues.

Convenience sampling: Selecting those sample that are most convenient or those immediately available.

Voluntary Sampling: Sample self-selecting. This is when individuals come forward voluntarily in response to an appeal from the researcher.

Snow ball sampling: This has to do with building up a sample through informants. You start with one person who then suggests another and so on.

Event Sampling: This involves using the opportunity presented by a particular

event e.g. a conference for sampling.

Time Sampling: This involves recognizing that different times or days of the week or year may be significant and sampling carried out at this period of the day



SAMPLE SIZE/ SAMPLE ADEQUACY

Sample size depend on research type

Use of Sample percentages (Newman 2000)

Descriptive research – at least 10% when the population is large and 20% or more when the population is small

Correlational research – at least 30 subjects

Experimental research – 30 subjects per group

Experimental with tight experimental control – 15 subjects per

group Causal comparative – at least 30 subjects per group

Sample Size Formula

Determination of sample size is one of the problems students face when writing research project. Determination of sample size precedes the construction of instruments. Miscalculation of sample size will lead to wrong result therefore the conclusion reached after analysis cannot be generalized to the entire population under study.

Taro Yamane formula has been found to be most reliable and effective way of calculating sample size. Yamane was a mathematician and statistician who developed statistical formula for calculating sample size based on the population under study so that inferences and conclusion reached after survey can be generalized to the entire population.

Sample size formula (Slovin, 1960; Taro Yamane,

$$1967) n = \frac{N}{1 + N(e)^2}$$

n = Sample size

N = Population Size

e = desired margin of error (Usually 0.05 for Social researches)

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NB: Other sample size formulae available

Use of Sample Size Table

The use of sample size table such as developed by Krejaie & Morgan (1970)

DESIGNING AND VALIDATING RESEARCH INSTRUMENTS

RESEARCH INSTRUMENTS (TOOLS)

The instruments for collecting research data are:

1. Observation
2. Interview
3. Document Analysis
4. Questionnaires/recording schedule
5. Rating scale/opinionnaire/attitude scale
6. Psychological test
7. Inventory/self-reporting scale.

The choice of research instrument depend on the nature of the study

Direct Observation

In the strict sense, observation means the use of the eyes rather than the ears and the voice. The concise oxford dictionary defines it as accurate watching and noting of phenomena as they occur with regard to the cause and effect or mutual relations. Observation when used in research gives a more reliable and valid results

The accuracy of information can be improved by the following procedures.

- (a) Observation should be planned
- (b) A written record should be made directly after an observation
- (c) Interpretation should follow the collection of data

Interview

Interview is a process that involves at least two people in a face-to-face oral discussion with the aim of wanting to get information one from the other. The two

people are **interviewer** and **interviewee** (candidate) respectively.



There are two main types of interview in use viz: **structured** and **unstructured**

interviews. Interview could also be conducted through the telephone.

Structured interview is one in which the content and procedure of the interview are organized in advance of the interview. Questions and alternative answers permitted are pre- determined, arranged in logical sequence and rigidly followed with all respondents. Interviewer may however deviate from the set questions for case of clarification of points.

The advantages of structured interview are that the approach is standardized and answer can easily be classified. Secondly, there is increased reliability as a result of restriction. It has disadvantage by the fact that the interviewee does not enjoy enough freedom to explain or expatiate on the responses made.

In **unstructured interview**, there is no format. The interviewer does not need to follow a laid down pattern of asking questions. He can thus go deep in the questioning exercise. The candidates enjoy ample opportunity.

The disadvantage of unstructured interview is that there is no sequence and vital questions may be forgotten. It does not lend itself readily to quantification.

On mode of interview, it may be conducted on a one to one basis or on a group basis (focus group). Other purposes for which interview may be used are admission, employment, promotion, scholarship award and leadership roles.

Precautions to be taken when conducting interviews are:

- Generate friendly situation.
- Do not argue with the interviewee.
- Do not show signs of content or disagreement.
- Give an accurate report devoid of bias.
- Observe high degree of confidentiality about the information in respect of the interviewee.

Document Analysis

Document analysis is a form of qualitative research in which documents are interpreted by the researcher to give voice and meaning around an assessment topic. Analyzing documents incorporates coding content into themes similar to

how focus group or interview transcripts are analyzed (Bowen, 2009).

Advantages of Document Analysis

1. Contain spontaneous data and therefore less time consuming. It requires data selection, instead of data collection.



2. Accessibility to information that would be difficult to get in any other way i.e readily available.
3. Make possible the collection of data over a longer period of time as well as larger samples that might be collected from questionnaires or interviews
4. Confessional documents may give us an insight into how people see things or how they want to present things.
5. Stability: Documents are non-reactive and stable. The investigator's presence does not alter what is being studied
6. Exactness: The inclusion of exact names, references, and details of events makes documents advantageous in the research process.
7. Relatively low cost, particularly when the documents are easily accessible and already located in your workplace, or on the internet.
8. Some types of document can be extremely detailed and yield much more information than you could hope to gain from a questionnaire or interview.

Disadvantages of Document Analysis

1. Documents are usually not designed with research in mind.
2. The information recorded may be idiosyncratic, bias or incomplete/can be too subjective.
3. Ethical principles may be contravened.

Criteria for Handling Documentary Sources

The general principles for handling documentary sources are not different from those of other applied and social sciences researches. Scott (199) in Mogalakwe (2006) formulated quality control criteria for handling documentary sources. These are:

- a. Authenticity
- b. Credibility
- c. Representativeness and
- d. Meaning

Authenticity refers to whether the evidence is genuine, and of reliable and

dependable origin. This is a fundamental criterion in any research. It is therefore the responsibility of the researcher to ensure that the document consulted is genuine and has integrity. There are many instances where documents may not be what they purport to be as documents such as wills, legal documents, diaries and letters can be forged or falsified. Some literary works may even



be attributed to authors who did not write them.

Documents should not be taken for granted thus placing enormous responsibility on researchers to satisfy themselves that the documents being analyzed are not forged and indeed what they purport to be. After establishing the authenticity of a document it is also important for the researcher to authenticate the authorship by verifying that the name inscribed on the document is that of the actual author.

Credibility refers to whether the evidence is free from error or distortion. Credibility concern the extent to which an observer is sincere in the choice of a point of view in an attempt to record an accurate account from the chosen standpoint. *Representativeness* refers to whether the evidence is typical of its kind while *meaning* refers to whether the evidence is clear and comprehensive. The ultimate purpose of examining documents is to arrive at an understanding of the meaning and significance of what the document contains.

Questionnaires

Questionnaire is a self-reporting system like the structured interview; but the responses given to questions are made in writing. Questionnaires are sometimes referred to as **recording schedule**.

As compared with interviews, the written questionnaire is typically more efficient and practical and allows for the use of large sample, standard instructions are given to all subjects and the personal appearance; mood or conduct of the investigator will not color the results. There are two types of questionnaires namely **structured (closed ended or form)** and **unstructured (open ended or form)**

Structured or closed form contains the questions and alternative answers. Unstructured questionnaire has the advantage of giving the respondent freedom to reveal their opinion and attitude. It has the disadvantage that the information generated is difficult to process and analyze.

The following should be avoided while constructing questionnaires.

- Avoid highbrow questions i.e. the use of high sound English to impress respondents. Use simple English.
- Avoid leading questions
- Avoid open ended questions or questions that are insufficiently specific.

Advantages of Questionnaires

- Can be used in the absence of the author.
- Can be conducted to many people at the same time.



- No psychological discomfort that may be experienced during face- to- face interaction.
- Can be used to obtain reliable information if well-constructed.

Disadvantages

- Misinterpretation of questions.
- Low return of questionnaires.

Rating Scale

Rating scale is a measuring device in which the assignment of numerals to the character of individual or behavior is made possible through the process of well-defined rules. The most widely used is the Likert scale for measuring attitude. The Likert scale is a five point level of agreement from strongly agree to strongly disagree. There are modified version of the scale today.

EXAMPLES

A		5	4	3	2	1
		SA	A	UN	D	SD
1	Jesus was born of a virgin without a human father					
2	There is a final day of judgment for all who have lived on earth					

B How often and which section of the Times Newspaper do you read?

	Always	Usually	Sometimes	Seldom	Never
1. State and local news					
2. Sports					
3. Advertising					

C How would you rate your Principal on the following traits?

		Excellent	V. Good	Good	Fair	Poor
1	Competence					
2	Friendliness					

3	Helpful					
4	Ability to inspire					

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Psychological Tests

A psychological test is an instrument designed to describe and measure a sample of certain aspect of human behavior. Test may be used to compare the behavior of two or more persons at a particular time or of one person or more at different times.

Psychological tests which are quantified by numerical scores yield objective and standardized description of behaviors. Psychological tests may be classified in terms of their purpose as achievement, aptitude, intelligence etc.

Inventories (Self reporting scale)

In inventories, the individual checks responses to certain questions or statements. Two main types are **interest and personality inventories**. Interest inventory is a measure of type of activities that an individual has a tendency to like and to choose while personality test measures certain personality traits or tendencies.

VALIDATION OF RESEARCH INSTRUMENT

It is necessary that research tools used are of very good quality. Two essential qualities of a good research instrument are **validity** and **reliability**.

Validity

The most important consideration is validity, which pertains to the question “what does this instrument measure?” Validity is the extent to which an instrument measures what it is supposed to measure. The face and content validity are commonly used in research.

Face Validity.

This refers to how reasonable the instrument appears as an instrument of the particular attribute intended. In other words, face validity is the extent to which one is able to accept on a mere superficial inspection of the instrument that it is actually an instrument of what it is intended to be. Face validity is not a serious form of validity.

Content Validity

This means the extent to which the content of the instrument correspond to those of the subject matter and their associated behavioral outcome. Content

validity is measured by analyzing the topical contents and the concepts to be measured. The closer the instrument contents are to those of the subject matter the higher is the content validity of the instrument.

Other forms of validity are criterion related validity, predictive validity, concurrent validity, congruent validity, constructs validity.

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Sources of Invalidity

- Unreliability
- Response sets - psychological orientation or bias towards answering in a particular way:
 - Acquiescence: tendency to agree, i.e. say "Yes".
 - Social desirability: tendency to portray self in a positive light.
 - Faking bad: The fake bad response style is defined as any conscious or unconscious attempt to produce item-- endorsement representing a more exaggerated symptomatology than is actually experienced by the respondent, with components of un-believability and an unrealistic negative impression.
- Bias
 - Cultural bias: When psychological construct do not have the same meaning based on culture. Actual content (face) validity may be different for different cultures.
 - Gender bias may also be possible.
 - Test Bias

Reliability

An instrument is said to be reliable if it measures consistently under varying conditions and at different times. Reliability is measured by means of statistics known as coefficient of reliability.

The main methods of assessing reliability are:

- (i) Test- retest method.
- (ii) Equivalent- forms method.
- (iii) Split- half method
- (iv) Scorer reliability.

Test- retest method.

To estimate reliability by means of test- retest method, the same instrument is administered to the same set of respondents with a given time interval between the two

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administrations. The resulting scores are correlated, and this correlation coefficient provides a measure of stability that is it indicates how stable the instrument results are over the given period of time. Today, an instrument could be administered once, coded and reliability determined by means of computer using the Cronbach's Alpha Reliability Method **Equivalent- forms method**

Estimating reliability by means of the equivalent- forms methods involves the use of two different but equivalent forms of the instrument (also called parallel or alternate forms). The two forms are administered to the same group of respondents in close succession and the resulting scores are correlated. The correlation coefficient provides a measure of equivalence. Thus it indicates the degree to which both forms of the instrument are measuring the same aspect of behavior.

Spilt- Half method

The reliability of an instrument can be estimated from a single administration of the instrument. The instrument is administered to a group of respondents in the usual manner and then divides in half for scoring purposes. To split the instrument in halves that are more equivalent, the usual procedure is to score the even-numbered items and the odd-numbered items separately. This provides two scores for each respondent, which when correlated provide a measure of internal consistency. This coefficient indicates the degree to which equivalent results are obtained from the two halves of the instrument.

To estimate the reliability based on the full-length test, the Spearman-Brown formula is usually applied. This formula is given as:

$$\text{Reliability of full test} = \frac{2 \times r}{1 + r}$$

Where r = Reliability of half test.

The simplicity of the formula can be seen in the following example where the correlation coefficient between the two halves of a test is 0.60

$$\text{Reliability of} = \frac{2 \times 0.60}{1 + 0.60} = \frac{1.20}{1.60}$$

$$\frac{1 + 0.60}{1.60} = 0.75$$

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Kuder- Richardson Method

Another method of estimating the reliability of a test from a single administration is by means of formulae such as those developed by Kuder and Richardson. These formulae provide a measure of internal consistency but they do not require splitting the test in half for scoring purposes.

A simpler formula for this purpose is the Kuder- Richardson formula 21 which can be applied to the results of any test that has been scored on the basis of the number of correct answers. This method is only suitable for dichotomized items e.g Yes/No, Agree/Disagree

Modified version of the

formula Reliability estimate

(KR 21)

$$= \frac{K}{K-1} \left(1 - \frac{M(K-M)}{KS^2} \right)$$

Where

K= the number of items in the

test. M= the mean of the test

scores.

S= the standard deviation of the test scores.

Scorer Reliability

This measures the extent to which different markers (inter-scorer) or the same marker (intra scorer) on different occasions will award the same score to the same answer. The marks awarded on different occasions by different markers or the same marker to the same set of testees is correlated to obtain a measure of scorer reliability.

RESEARCH DATA ANALYSIS

Research data analysis is the most important aspect of research. The issue of statistical data analysis is more of a specialized field, but students carrying out research need at least a rudimentary aspect of data analysis. Following are hints on qualitative data analysis, quantitative data analysis and the use of computer in data

analysis.

Qualitative Data Analysis

In qualitative research, the focus is on in-depth interviews, observations and document analysis. In general, qualitative research interprets data without numerical analysis, however some qualitative research include limited quantification e.g counting the number of occurrences of an event.



There are three processes in qualitative data analysis namely data organization, data description and data interpretation. The first which is data organization is critical due to the often voluminous data collected in quantitative studies. The second step is the description of the data, the setting in which they were collected and the participants in the study. Finally, the data must be interpreted.

Quantitative Data Analysis

Quantitative research requires statistical testing. There are four major statistical testing that could be used by students in their data analysis. These are:

- i. Test of Significance
- ii. Test of central tendency
- iii. Test of variation and
- iv. Parametric and non-parametric tests

In addition to using the statistical methods above, the researcher could use charts, graph, diagrams, models and other figures to drive home their points to provide clear understanding of their solution to the problem. For orderliness and systematic solution of the research problem, the research hypothesis or questions could be tackled one after the other and appropriate test used for each of them

Computer Data Analysis

The electronic digital computer is one of the most versatile developments of the technological age. To the researcher, the use of computer to analyze complex data has made complicated research design practical. Performing calculations almost at the speed of light, the computer has become one of the most useful tools in the physical and behavioral sciences as well as humanities.

The steps in using computer to calculate statistical analysis are;

1. Data organizing and coding
2. Storing the data in the computer
3. Labeling the variables
4. Selecting of appropriate descriptive and inferential statistics
5. Selection of appropriate programmes for described statistics
6. Execution of the computer

programme Two main statistical packages are

SPSS – Statistical Packages for Social Sciences

SAS – Statistical Analysis System



ETHICAL CONSIDERATIONS IN RESEARCH

Informed Consent. Potential research participants must be provided with information that enables them to make an informed decision as to whether they want to participate in the research study

Informed Consent from parents of Participants. Informed consent must be obtained from parents or guardians of minors.

Do not withhold information from participants about the nature and/or purpose of the study

Freedom to Withdraw: Participants must be informed that they are free to withdraw from the study at any time without penalty.

Confidentiality and Anonymity

Confidentiality is a basic requirement in all studies. It means that the researcher agrees not to reveal the identity of the participant to anyone other than the researcher and his or her staff.

A stronger and even better condition (if it can be met) is called **anonymity**. Anonymity means that the identity of the participant is not known by anyone in the study, including the researcher. An example would be where the researcher has a large group of people fill out a questionnaire but NOT write their names on it. In this way, the researcher ends up with data, but no names

RESEARCH PROPOSAL

Framework of the Research Proposal

A research proposal is a written document that includes the following information:

- Introduction
- Literature Review
- Research Methodology

The introduction part introduces your research idea, establish its importance and explain its significance. The literature review is an exploration of what other researchers have done in the area of study, while the research methodology explain in details how the research will be conducted. It provides a written description of

the specific actions, plan, or strategy you will take to answer your research questions.

The purpose of your proposal is to sell your idea by showing you have thought it through



very carefully and have planned a good research study.

The research proposal can serve many useful functions.

- The most important is that it helps you to think out the research project you are about to undertake and predict any difficulties that might arise.
- For those who are not sure of what their focus will be, the research proposal can be a space to explore options -- perhaps with one proposal for each potential topic. These can be more easily compared and evaluated than when they are still just ideas in one's head.
- Research proposals can be effective starting points to discuss projects with your professors.
- A professor who is initially skeptical about a project may be able to imagine it more easily after reading a well written research proposal (this does not mean that he or she will approve the topic, especially if there are significant potential difficulties that you have not considered).
- Once you have begun your research project, a research proposal can help you to remain on track -- and can also remind you why you started this project in the first place

The format for the writing a research proposal may vary from one department to another and as a result research students need to check with the department.

WRITING THE FINAL RESEARCH REPORT

A good research with many meaningful findings may lose its value and be thrown into the dust-bin if the writing of the research report is badly done. A good research report writing entails hard work and patience.

Researchers can adopt any format for writing the report but should, essentially reflect the research process. A sample of an outline of search report is presented below:

Preliminary Pages

Title Page

Certification

Dedication

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Acknowledgement

Table of

Contents List of

Tables

List of Figures

List of

Appendices

Abstract

CHAPTER ONE: INTRODUCTION

Background to the

Study Statement of the

problem Purpose of the

study Objectives of the

Study

Research

Questions/Hypotheses

Significance of the study

Scope and Delimitations of the Study

Conceptual/Theoretical Framework

Operational Definitions of Terms

CHAPTER TWO: REVIEW OF RELATED

LITERATURE CHAPTER III: RESEARCH

METHODOLOGY

Research Approach

Research Design

Population of the Study

Sample and Sampling Technique

Instrumentation

Validation of

Instrument Method of

Data Collection Method

of Data Analysis Ethical

Consideration



CHAPTER FOUR: DATA ANALYSIS AND INTERPRETATION

CHAPTER FIVE: SUMMARY, CONCLUSION AND RECOMMENDATION

REFERENCES

APPENDICES

WRITING RESEARCH REPORTS

Writing the main report of a study:

One of the most important aspects of any research study is communication of the results to others. This important aspect can be accomplished through a well written report. Most reports start with the **title**. The title is the label given to the report. It should be written in such a way that the reader can easily tell what the study is all about, even without reading the whole report. The title should not be more than 20 substantive words. After the title comes the **abstract**. The Abstract should contain important ideas of the study. The abstract is a brief summary of the whole study containing main steps in the study, statement of purpose, the types of subjects, brief statement of the design, data analysis, result, and findings. Usually the abstract should not be more than 350 words.

Chapter One: Introduction

This chapter usually includes the introduction where the reader should be properly introduced to the problem. After the introduction comes the statement about the background to the study and other sub-heading as spelt out above

Chapter Two: Review of Related Literature.

Chapter two deals mainly with review of related literature highlighting previous research work in related studies. Review of previous studies should not be limited only to local studies. It should be international in scope.

Chapter Three: Research Design and Procedure

Here, the researcher should clearly explain step by step the nature of design used, the sample, the instrument, and the procedure of the study.

Chapter Four: Data Analysis and Interpretation

Data could be analyzed quantitatively or qualitatively. In quantitative data analysis, the important thing under presentation of results is that all tables, charts, graphs etc. should be clearly labeled and explained so that readers may understand the report. Results are to be presented in line with the research questions/hypothesis.

Chapter 5: Summary, Conclusion and Recommendation

In this chapter the researcher summarizes his research procedure, results and discussions; he then draws concrete conclusions from his findings and makes scholarly recommendation based on his findings.

References, Appendices and Bibliography

A. References

The final state on reporting is referencing. References are those works or studies which the research cited in his report. This does not include all other works he consulted but did not cite. If however, the researcher wants to include the list of the works of studies he consulted but not cited in addition to the one cited, then he is not writing a reference, but a bibliography. :

Usually references and bibliographies are listed in alphabetical order. First the authors family name, initials, year and title of the work. If the cited work is a journal, the journal name is underlined, followed by the volume, number and pages. If the cited work is a book or other materials, the title is underlined, followed by the country/city of publication and publishers. References should be indicated clearly enough for the reader to get at it if he so wishes.

Three commonly used referencing styles are:

American Psychological Association (APA) – Education and Social

Sciences Modern Language Association (MLA) - Languages

Kate Turabian Style – Theology.

NOTE: For details and guidance, search for APA referencing style online.



The APA Referencing Style (Sample)

Books (One author)

Kmoisar, L. (1991). *The new feminism*. New York: Franklin Watts.

Books (Two to three authors)

Strunk, W., Jr., & White, E.B. (1979). *The elements of style* (3rded.).

New York: Macmillan.

Journal Article (One author)

Maki, R.H.(1982). Categorization effects which occur in comparative judgment tasks. *Memory & Cognition*, 10, 252-264.

Journal Article (Two authors)

Atkinson, R.C., & Shiffrin, R.M. (1971). The control of short-term memory. *Scientific American*, 225, 82-90

Magazine Article (No author)

The blood business (1992, September 11). *Time*, 97, 47-48.

Magazine Article (One author)

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Note: For details and guidance, search for APA referencing style on the internet.

b. Appendices

There are various items that a researcher does not want to include in the body of the report, such things as raw data, transmittal letter, certain maps and charts, detailed calculations, instruments etc are put under an appendix. Appendices are put after the final chapter and references.

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Any other Research reading materials

